Hanlin Hao

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Education

- **University of Bonn** *Oct. 2024 – Present*
  M.Sc. in Mathematics Bonn, Germany
- **Wuhan University** *Sept. 2020 – Jun. 2024*
  B.Sc. in Mathematics and Applied Mathematics, GPA: 3.80/4.00 Wuhan, China

Research Interests

Algebraic Geometry, Complex Geometry, Moduli Spaces, K-theory and Hall Algebras, Derived Categories, Representation Theory, and Symplectic Geometry.

Research Experience

- **Master's Research Project** *Oct. 2026 – Present*
  K-theoretic Hall algebras of torsion sheaves on curves University of Bonn
  Supervisor: Tudor Pădurariu.
  - Established a shuffle description of Hall multiplication on the localized K-theory of moduli stacks of torsion sheaves on smooth curves.
  - Currently investigating the derived categories of these stacks using window categories, with the aim of understanding their semiorthogonal decompositions.
- **Undergraduate Thesis** *Jan. 2024 – May 2024*
  Complex Tori and Line Bundles Wuhan University

Teaching Experience

- **Tutor, Representation Theory I** *Summer Semester 2026*
  University of Bonn Bonn, Germany
- **Tutor, Representation Theory II** *Winter Semester 2026/27*
  University of Bonn Bonn, Germany

Employment

- **Student Assistant (Library)** *Oct. 2026 – Present*
  University of Bonn Bonn, Germany

Conferences, Workshops and Schools

- **Essen School on Exponential Motives** *Sept. 20–25, 2026*
  School on exponential motives Essen, Germany
- **Two Centuries of Moduli** *Sept. 6–11, 2026*
  Conference on moduli spaces Berlin, Germany
- **Early Number Theory Researchers Workshop 2025** *2025*
  Early-career workshop in number theory Paderborn, Germany
- **MAPSS 2025** *Aug. 15–24, 2025*
  Mathematical Physics Summer School
- **Cambridge Winter Programme in Theoretical Neuroscience** *Jan. 2021 – Mar. 2021*
  Balanced Network Models of Cortical Circuits

Skills and Languages

Languages: Chinese (native); English (C1, IELTS 7.0).

Programming: Python, C, MATLAB.

Typesetting: LaTeX.